



ZIMBABWE

MINISTRY OF PRIMARY AND SECONDARY EDUCATION

BIOLOGY SYLLABUS

FORMS 3 – 4

2024 - 2030

Curriculum Development Unit
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HARARE

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BIOLOGY SYLLABUS F3-4 2024 - 2030

TABLE OF CONTENTS

Table of Contents

PREAMBLE	4
INTRODUCTION.....	4
RATIONALE.....	4
SUMMARY OF CONTENT.....	4
ASSUMPTIONS	4
1.5 CROSS- CUTTING THEMES	5
2.0 PRESENTATION OF SYLLABUS.....	5
4.0 OBJECTIVES	6
5.0 METHODOLOGY AND TIME ALLOCATION	6
5.1 METHODOLOGY	6
5.1 TIME ALLOCATION	7
6.0 TOPICS	7
7.0 SCOPE AND SEQUENCE.....	8
8.0 COMPETENCY MATRIX	1
FORM 3.....	1
FORM 4.....	41
9.0 ASSESSMENT.....	1
9.1 Assessment Objectives.....	1
9.2 Assessment Model.....	1
9.3 Scheme of Assessment.....	2
9.4 Specification Grid.....	5

1.0 PREAMBLE

1.1 INTRODUCTION

The Biology Syllabus is designed for learners in Forms 3-4. The learners are expected to acquire theory and practical skills as well as develop socially, emotionally, physically, and cognitively. The syllabus aims at balancing knowledge, understanding, and skills in order to enable candidates to become effective learners and to provide a firm foundation for such careers as in Medicine, Food Technology, Biotechnology, and Environmental Science.

1.2 RATIONALE

This Heritage Based Biology syllabus encourages learners to employ biological skills in solving real life problems and also emphasizes the link between human activities and the environment. Learners acquire knowledge and skills of inquiry that help them to examine critically issues that arise in their own lives and in the public domain, to contribute to debate and to take part in making decisions about their own health and well-being and that of the society. The learners will be assessed through a continuous assessment system in the form of project-based assessments, hands on experiences and demonstrations.

1.3 SUMMARY OF CONTENT

The content covered by this syllabus includes theory and practical skills in the broad areas of Biology such as: Cytology, Anatomy, Physiology, Genetics, Ecology, Systematics, Health and Diseases.

1.4 ASSUMPTIONS

The Heritage-Based Biology syllabus for Zimbabwe has taken a deliberate consideration of several assumptions critical for socio-economic transformation. The assumptions are based on the context of Zimbabwe's heritage, educational system, societal needs and aspirations. It therefore becomes critical to consider that learners:

- are exposed to scientific experiences
- live in diverse social contexts
- use technological devices
- are conscious of their environment
- are aware of their obligation towards health and well-being

The general assumption is that a Heritage-Based Biology syllabus can effectively integrate science and technology education, fostering a deeper understanding of scientific concepts, technological innovations and their relationship with Zimbabwe's heritage

The syllabus assumes that learners have:

- acquired skills in handling apparatus
- learnt about the hazards and safety precautions during experiments
- knowledge on the basic concepts of biological sciences
- developed an awareness and interest in the conservation of the environment
- a basic understanding of health and disease issues

1.5 CROSS- CUTTING THEMES

In order to foster competency development for further studies, life and work, the following cross-cutting priorities have been taken into consideration:

- Gender equity
- Environmental management
- ICT
- Children's Rights and Responsibilities
- Disaster Risk Management
- Health and wellbeing
- Climate Change
- Entrepreneurship

2.0 PRESENTATION OF SYLLABUS

The Forms 3 - 4 Heritage-Based Biology Syllabus is presented as a stand-alone syllabus which covers the major branches of Biology. It consists of the preamble, aims, objectives, methodology, scope and sequence, competency matrix and assessment.

3.0 AIMS

The aims are to help learners to:

- 3.1 appreciate the importance of biology for sustainable socio-economic development of the country
- 3.2 develop biological practical skills, accuracy, objectivity, integrity and enquiry
- 3.3 develop good practices for health and safety
- 3.4 apply scientific method in Biology, other disciplines and in solving everyday life experiences
- 3.5 recognise that the study and practice of biology are inter-related and are subject to economic, technological, social, political, ethical and cultural influences

3.6 develop interest and participate in caring for the local and global environment

4.0 OBJECTIVES

Learners should be able to:

- 4.1 demonstrate knowledge of biological terms, laws, facts, concepts, theories and phenomena
- 4.2 select and use appropriate technological instruments to collect and analyse data
- 4.3 plan and conduct experiments in which they may:
 - identify a problem
 - formulate research questions and hypotheses
 - predict results of investigations base upon prior data
 - identify variables and describe the relationship between them
 - plan procedures to control independent variables
 - collect data
 - select the appropriate format such as: graph, chart, diagram and use it to summarise the data obtained
 - analyse data, check it for accuracy and construct reasonable conclusions
 - prepare written and oral reports of investigations
- 4.4 draw biological diagrams in two dimensions
- 4.5 carry out simple scientific calculations
- 4.6 translate information from one form to another
- 4.7 make logical decisions based on the examination of evidence and arguments
- 4.8 communicate biological information logically and concisely
- 4.9 apply biological principles in solving problems and in understanding new situations
- 4.10 identify the practical constraints affecting biological investigations
- 4.11 use biological principles, methods and techniques in value addition
- 4.12 apply appropriate methods of recycling and waste disposal
- 4.13 explain the effects of technological applications on the environment and the organisms within it
- 4.14 interpret the relationship between living organisms and between living organisms and their environment

5.0 METHODOLOGY AND TIME ALLOCATION

5.1 METHODOLOGY

The syllabus is based upon interactive, multi-sensory, learner centred and practical approaches. Principles of individuality, team work, wholeness and stimulation must be applied to enhance the learning – teaching process. The learners should be allowed to apply their experiences, knowledge, skills and attitudes in the learning of the subject. The following are the suggested methods:

- Experimentation
- Discovery method
- Demonstrations
- Problem solving
- Discussions
- Visual tactile
- e-learning
- Group work
- Educational tours
- Resource person(s)
- Project based learning
- Case studies
- Observations
- Simulations

5.1 TIME ALLOCATION

A time allocation of eight (8) periods of 40 minutes per week is recommended. Double periods are also recommended

6.0 TOPICS

The Heritage -Based Forms 3 -4 syllabus consists of thirteen compulsory topics listed below:

- 6.1 Introduction to Biology
- 6.2 Chemicals of Life
- 6.3 Cells and Cellular Activities
- 6.4 Enzymes
- 6.5 Plant Science
- 6.6 Animal Science
- 6.7 Respiration
- 6.8 Microbiology and Biotechnology
- 6.9 Genetics
- 6.10 Biodiversity
- 6.11 Ecosystems
- 6.12 Health and Disease

7.0 SCOPE AND SEQUENCE

TOPIC	FORM 3	FORM 4
6.1 Introduction to Biology	• Branches of biology • Safety in the lab • Careers	• Safety (Labels)
6.2 Chemicals of Life	• Water • Carbohydrates • Proteins • Fats • Nucleic acids	• Carbohydrates • Proteins • Fats • Nucleic acids
6.3 Cells and Cellular Activities	• Structures of plant and animal cells • Cell specialization • Cell transport	
6.4 Enzymes	• Nature and properties of enzymes • Mode of action	• Industrial application of enzymes
6.5 Plant Science	• Nutrition • Productivity • Transport	• Coordination and response • Reproduction
6.6 Animal Science	• Nutrition • Gaseous exchange • Transport • Sexual reproduction in Humans	• Productivity • Homeostasis • Coordination and response • Skeletal system
6.7 Respiration	• Types	• Application
6.8 Microbiology and Biotechnology	• Types of microorganisms • Characteristics • Economic importance of microorganisms	• Recombinant gene technology
6.9 Genetics	• Genes and chromosomes • Monohybrid inheritance • Mutations	• Variation • Selection
6.10 Biodiversity	• Classification	• Threats
6.11 Ecosystems	• Natural and artificial	• Management

6.12 Health and Diseases	• Health • Diseases (Infectious and non-infectious)	• Drug use and abuse
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BIOLOGY SYLLABUS F3-4 2024 - 2030

7.0 COMPETENCY MATRIX**FORM 3****7.1 TOPIC 1 INTRODUCTION TO BIOLOGY**

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.1.1 Branches of Biology	- define Biology - identify various branches of biology	- Definition of Biology - Branches - Cytology - Anatomy - Physiology - Ecology - Genetics - Microbiology - Zoology - Botany	- Discussing the meaning of the term Biology - Bio-life - Logy – study - Discussing the various branches of biology	- Relevant reference materials - Recommended textbooks - ICT tools
7.1.2 Careers	list biology related careers	- Careers such as: - Medicine - Research scientists	Researching on careers related to	- ICT tools - Career guidance booklet

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
		- Ecologists - Conservationists - Dieticians	various branches of biology	• Resource persons
7.1.3 Safety in the Biology Laboratory	• identify causes of accidents in the lab • demonstrate proper handling of apparatus • outline laboratory safety rule • perform fire drills • memorise the emergency numbers • make a fire guard around the lab • demonstrate use of the First Aid Kits	• Causes of accidents in the bio lab - Fire - Fumes - Acids and strong bases - Handling of micro-organisms - Improper handling of apparatus - Electricity • Laboratory safety rules - Fire drills - Emergency call for help	• Identifying causes of accidents in the lab • Listing accidents that may occur during experiments • Demonstrating proper handling of apparatus • Discussing electrical hazards • Discussing laboratory safety rules • Performing mock fire drills • Demonstrating emergency call • Preparing a fire guard around the lab	• Recommended textbooks and materials • Charts showing electrical hazards • ICT tools • Fire extinguishers • Fire blankets • Sand buckets • Fire alert bells • Fume cupboards • Goggles and masks • Gloves • First Aid Kit

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
			Demonstrating use of the contents of the First Aid Kit	

7.2 TOPIC 2 CHEMICALS OF LIFE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.2.1 Water	- state the constituent elements of water - describe the chemical structure of water - relate the properties of water to its uses in living organisms and as a habitat for living organisms	- Constituent elements - Chemical structure - Properties and uses	- Naming the constituents of water - Constructing the model of a water molecule - Demonstrating the unusual behavior of water when it freezes - Relating the properties of water to its uses in living organisms	- Liquid and frozen water models of a water molecule - ICT tools - Recommended textbooks

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.2.3 Carbohydrates, Proteins, Fats and Nucleic Acids	- state the constituent elements of carbohydrates, proteins, fats and nucleic acids - name the sub-units of carbohydrates, proteins, fats and nucleic acids - perform tests to identify reducing sugars, non-reducing sugars, proteins and fats	- Constituent elements - Sub-units of carbohydrates (starch, cellulose, glycogen), proteins, fats and nucleic acids - Identification of reducing sugars, non-reducing sugars, starch, proteins and fats	- Identifying the constituent elements of carbohydrates, proteins, fats and nucleic acids - Identifying - Glucose as sub units of starch, cellulose, glycogen - Amino acids as sub-units of proteins - Fatty acids and glycerol as sub-units of fats - Nucleotides as sub units of nucleic acids - Carrying out the following identification tests: - Iodine test for starch - Benedicts test for reducing sugar - Biuret test for proteins - Emulsion/spot test for fats	- Recommended textbooks - Iodine solution - Benedicts solution - Biuret solution/sodium hydroxide and copper sulphate - Hydrochloric acid - Sodium hydrogen carbonate - Ethanol - Water - Filter paper

7.3 TOPIC 3 CELLS AND CELLULAR ACTIVITIES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.3.1 Plant and Animal Cells	• define cells • identify parts of the cells • compare the structures of plant and animal cells • outline the functions of cell organelles	• Structural and functional units • Plant and animal cell structure - Nucleus - Cytoplasm - Cell membrane - Vacuole - Cell wall - Mitochondria - Chloroplasts	• Defining cells • Observing cells under the microscope and bioviewers • Drawing and labelling plant and animal cells • Comparing the structures of plant and animal cells	• Microscope • Biosets and bioviewers • Recommended textbooks • Print media

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
			Discussing functions of cell organelles	
7.3.2 Cell Specialisation	- Identify specialized cells - Relate cell structure to function	- Palisade – photosynthesis - Root hair – absorption - Neurone – transmission of impulse - Red blood cell – transport of oxygen - White blood cell – immunity - Muscle cell – contraction - Sperm – fertilization and motility - Ovum – fertilization and food reserve	- Relating cell structure to function - Examining specialized cells using bioviewers, microscopes, photomicrographs	- Biosets - Bioviewers - Photomicrographs - Print media - Microscopes
7.3.2 Cellular Transport	- define diffusion - investigate factors that affect the rate of diffusion - define osmosis - describe active transport	- Diffusion - Factors affecting diffusion rate: - Surface area/volume ratio, temperature and particle size - Osmosis - Turgor	- Demonstrating diffusion as movement of particles in gases, liquids and solids - Describing diffusion - Demonstrating osmosis	- Perfume - Potassium Permanganate - Potatoes - Visking tubing - Egg shell membrane - Onion skin - Microscope

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
		- Plasmolysis • Active transport	• Movement against concentration gradient	

7.4 TOPIC 4 ENZYMES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.4.1 Nature and properties of enzymes	• define enzyme • explain the properties of enzymes • explain enzyme action in terms of the lock and key hypothesis • investigate and describe the effects of temperature and pH on enzyme activity	• Biological catalysts • Protein nature and substrate specificity • Mode of action (The lock and key hypothesis)	• Discussing the concept of enzyme • Carrying out experiments to investigate properties of enzymes such as: amylase from germinating rapoko seedling and catalase from fresh plant and animal tissue (N.B. No	• Recommended textbooks • Rapoko or any other grain • Fresh tissues from plants and domestic animals • ICT tools • Models

			tissues from human or wild animals) Using models and animations to illustrate the lock and key concept	
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7.5 TOPIC 5 PLANT SCIENCE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.5.1 Nutrition	- define photosynthesis - State the word equation for photosynthesis - Investigate conditions necessary for photosynthesis - explain the importance of	- Photosynthesis - Word equation - Conditions necessary chlorophyll, light, carbon dioxide - Fate of end products - Importance of photosynthesis	- Describing the process of photosynthesis - Carrying out controlled experiments to show conditions necessary for photosynthesis	- Apparatus for testing for starch in leaves - Potted plants - Iodine solution - Hand lenses - Microscopes - Bioviewers - Lime water/bicarbonate indicator

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	photosynthesis in an ecosystem - investigate factors affecting the rate of photosynthesis - relate the structure of the leaf to its function in photosynthesis - investigate gaseous exchange in plants	- Factors affecting the rate of photosynthesis (light intensity; carbon dioxide concentration and temperature - Adaption of the leaf to photosynthesis - Gaseous exchange in plants (leaves)	- Discussing the fate of end products - Discussing the importance of photosynthesis - Experimenting on the effect of these factors - Observing the external structure of leaf - Observing prepared slides of leaf internal structure - Carrying out experiments to demonstrate gaseous in plants using lime water/bicarbonate indicator	Live potted plants
	describe the functions of nitrogen, phosphorus and	- Mineral Nutrients - Nitrogen (N) – protein synthesis and	Discussing the functions of nitrogen, phosphorus and	- Recommended textbooks - ICT tools

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	potassium on plant growth • describe the effects of deficiencies of nitrogen, phosphorus and potassium	chlorophyll formation - Phosphorous (P) – synthesis of ATP and nucleic acids - Potassium (K) – ion and osmotic balance	potassium on plant growth	
	• describe the effects of deficiencies of nitrogen, phosphorus and potassium	• Deficiency symptoms of Nitrogen, phosphorus and potassium - N – stunted growth and chlorosis - P – Stunted root growth and purplish leaf colouring - K – yellow and brown leaf margins and premature death (poor	• Carrying out culture experiments	• Culture solutions • Plants

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
		flowering and fruiting)		
7.5.2 Productivity	- define biomass - define productivity - explain factors affecting productivity	- Biomass – organic content - Increase in biomass overtime - Factors affecting productivity - Light - Mineral salts - Temperature - Water availability - Pests and diseases	- Carrying out experiments to measure biomass - Discussing productivity in terms of increase in biomass overtime and factors affecting	- Balances - Ovens - Recommended textbooks - Print media
	- define pest - identify types of plant pests and diseases	- Pest as an organism which reduces productivity - Plant pests and diseases <u>Pests</u> - Tissue – eating and sap-sucking pests <u>Diseases</u>	- Field visit observing plants affected by various pests and diseases - Reporting back	- School grounds and neighboring fields - Preserved plant pests and affected plants

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
		- Bacterial wilt and fungal rust		
	- explain how pests and diseases affect productivity - describe methods of controlling pests and diseases - outline the advantages and disadvantages of each method	- Reduction of yields - Management, cultural, chemical and biological	- Discussing effects of plant pests and diseases on productivity - Discussing methods of controlling plant pests and diseases with specific reference to maize, cotton and tobacco	Resource person
7.5.3 Transport	- describe the general structure of a plant - distinguish between monocotyledonous and dicotyledonous plants	- Plant structure (external) - Shoot system – stem, leaves, flowers, fruits, seeds - Root system	- Educational tour to observe the general structure of plants - Comparing the structures of dicotyledonous and monocotyledonous plants	Hand lenses
	identify parts of the internal structure of a young	- Plant internal structure - Stem – Epidermis,	Observing prepared slides showing internal	- Microscopes - Bioviewers

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	dicotyledonous stem and root compare internal structure of dicotyledonous and monocotyledonous stems	cortex and vascular tissue (xylem and phloem) Root – Epidermis, cortex, vascular tissue and root hairs	structures of stem and root Making labelled drawings of observed structures	
	state the functions of xylem and phloem in a plant	- Xylem – support and transport - Phloem – transport	- Discussing the functions of xylem and phloem - Carrying out experiments using dyes to identify vascular tissue	- Dyes - Celery plants or spinach
	relate the structure of root hairs to their function	- Root hairs - Adaptations of root hairs to their functions - Large surface area for water and ion uptake - Diffusion, osmosis and active transport	- Observing root hairs from microscope slide - Drawing labelled diagrams of root hairs	- Microscope slides - Bio viewers

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	- define transpiration - describe the functions of transpiration	- Transpiration is the evaporation of water at the surfaces of the mesophyll cells followed by the loss of water vapour from the leaves through the stomata - Functions of transpiration - Water and ion movement in xylem - Cooling effect	Discussing the concept and functions of transpiration	- ICT tools - Recommended textbooks
	investigate factors affecting rate of transpiration	- Factors affecting rate of transpiration - Surface area - Stomata distribution - Temperature - Wind - Humidity - Light intensity	- Carrying out experiments to investigate factors affecting transpiration - Demonstrating transpiration using a potometer, balance and cobalt chloride paper	- Potometer - Balances - Cobalt chloride paper - Plant shoots
	describe adaptations of leaves to	Reduction of surface area	Observing leaves	- Leaves - Vaseline

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	minimize water loss	• Thickness of cuticle • Distribution of stomata • Presence of hairs	• Carrying out experiments to investigate the distribution and role of stomata in water loss	• Transparent plastic bags
	• describe how wilting occurs • define translocation • describe negative effects of bark ringing to a plant	• Wilting and its importance • Translocation as movement of organic solutes in phloem • Negative effects to the environment of using barks to make artefacts	• Observing wilting plants • Explaining wilting as excessive loss of water during transpiration • Explaining the experiment on bark ringing • Explaining the effects of bark ringing to the environment with specific reference to tree conservation	• Potted plants • Print media • Educational touring

7.6 TOPIC 6 ANIMAL SCIENCE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.6.1 Nutrition	- define balanced diet - list components of a balanced diet - Test food for presence of nutrients - list major sources of nutrients - describe the major function of nutrients	- Human nutrition - Balanced diet (a diet containing all essential nutrients in the right quantities and proportions for a health life) - Components of a balanced diet - Sources of nutrients - Functions of nutrients NB. On fats include synthesis of cell membranes	- Discussing balanced diet - Carrying out food tests - Discussing sources and functions of nutrients	- Recommended textbooks - Refer to topic 2

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	compare energy content of nutrients	Carbohydrates and fats	Carrying out experiments to measure energy in food	Carbohydrate and fat-rich foods
	relate food intake to energy expenditure, growth and health requirements	Balance between food intake, energy expenditure, growth and health requirements (Age, sex and activity)	Discussing the relationship between food intake to energy expenditure, growth and health requirements	- Recommended textbooks - Resource persons
	- define malnutrition - discuss causes and effects of malnutrition	- Malnutrition as under nourishment or over nourishment - Causes and effects of - Deficiency diseases - Obesity - Coronary heart disease	- Discussing malnutrition - Carrying out case studies on peoples' eating habits and common malnutrition problems in their community - Carrying out case study on nutritional values of traditional and locally available foods - Discussing effects of taking excess fats, carbohydrates and salt	- Recommended textbooks - ICT tools - Resource person

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	- identify the main regions in the human alimentary canal and associated organs - draw and label the alimentary canal - state functions of parts of the alimentary canal	- Human alimentary canal - Ingestion, digestion, absorption, assimilation and egestion	- Identifying and drawing parts of the human alimentary canal - Discussing functions of the parts of the alimentary canal	- Printed media - Recommended textbooks - Model
	- define a ruminant - explain rumination - draw and label the digestive system of a ruminant	- Ruminants - Digestive system	- Dissecting a ruminant - Identifying the different stomach chambers - Relating each stomach chamber to function - Drawing the digestive system	- A ruminant - Dissecting kit - Print media
	- define mechanical digestion - describe peristalsis - describe main functions and types of teeth	Mechanical Digestion as physical breakdown of food by teeth and muscles	- Discussing mechanical digestion - Describing peristalsis - Identifying incisors, canines, premolars and molars - Identifying the crown neck, root, enamel, dentine and pulp cavity	- Printed media - Model of a tooth

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	- describe the structure of a tooth - identify causes of dental decay			
	identify causes of dental decay	- Causes of dental decay - Action of bacteria on sugary food producing acids	- Describing causes of tooth decay - Experimenting on action of acid on tooth	- Printed media - Model of a tooth
	describe the proper care of teeth	- Care of teeth - Remove plaque - Neutralizing acid - Destroying bacteria	Describing different ways of preventing tooth decay	- Acid - Tooth or any calcium containing material - Tooth paste - Tooth brush - Flossy - Tooth pick - Salt - Ash/soda
	explain the importance of chemical digestion	- Chemical digestion - molecule size and solubility	- Discussing chemical digestion - Carrying out experiments with starch and amylase in visking	- Starch solution - Visking tubing - Amylase

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	- describe the function of amylase, protease and lipase - state the end products of digestion of starch, proteins and fats	- breakdown of starch, protein and fats - starch – glucose - proteins – amino acids - fats – fatty acids and glycerol	tubing as a model gut to illustrate chemical digestion and absorption (testing for glucose)	- Thermometer - Benedict solution
	explain the roles of hydrochloric acid, rennin and bile	- Hydrochloric acid – buffer, destruction of bacteria and hydrolysis - Rennin – clotting of milk - Bile – emulsification of fats	Carrying out experiments on hydrolysis, clotting of milk and emulsification of fats	Hydrochloric acid, bile, starch, fats and rennin
	- describe structural adaption of small intestines to absorption - describe absorption of end	Features of small intestines: length, folding, villi, thin lining, rich blood supply	- Analyzing features of the small intestine - Discussing absorption of end products of digestion and their transportation to the liver - Describing roles of the large intestine	- Printed media - Model of villi - Recommended textbooks

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	products of digestion - state the function of the hepatic portal vein - Identify the role of large intestines	- Diffusion and active uptake - Absorption of water, vitamin B formation and expulsion of faeces - Transportation of products of digestion to the liver		
	- define assimilation - outline the role of the liver in metabolism	- Use of end products of digestion - Liver as a storage and regulatory organ - Regulation of glucose and amino acids - Breakdown of alcohol and toxins - storage of fat, glycogen and vitamin A and D	- Discussing the fate of end products of digestion - Discussing the roles of the liver NB. Details of urea formation not required	- Recommended textbooks - ICT tools

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.6.3 Gaseous Exchange in Mammals	- identify parts of the respiratory system - state the functions of parts of the respiratory system - describe the mechanisms of breathing - describe the structure and role of the alveoli - describe factors which increase the efficiency of gaseous exchange	- Nasal passages, larynx, trachea, bronchi, bronchioles, alveoli and capillaries - Mechanisms of breathing - Structure and role of alveoli - factors which increase the efficiency of gaseous exchange	- Identifying parts using models - Discussing functions of parts of the respiratory system - Demonstrating breathing mechanisms using models and animations - Discussing factors which increase efficiency of gaseous exchange	- Use of models - Recommended textbooks - Printed media - ICT tools
	state the difference between inhaled and exhaled air	Changes in percentage volume of carbon dioxide, oxygen and water vapour	Carrying out experiments to show that exhaled air contains a higher proportion of carbon dioxide	Lime water/ Bicarbonate Indicator

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	explain the effects of physical activity on breathing and blood circulation	- Depth and rate of breathing - Pulse rate	Measuring of pulse rate and breathing rates before and after exercises	Stop watch
	describe some effects of smoking on respiratory system	- Short term effects - Long term effects	Evaluating statistical data linking diseases and smoking	- ICT tools - Print media - Resource person
7.6.8 Transport in Mammals	describe the human circulatory system	heart, vessels, blood and valves	describing components of the circulatory system	- Printed media - Models - ICT tools
	state functions of the circulatory system	Transport of materials and distribution of heat	Discussing transportation of materials and distribution of heat	Recommended textbooks
	- identify parts of the heart - describe the cardiac cycle - explain the importance of heart valves	- Parts of the heart - Cardiac cycle NB. Mentioning of diastole and systole not necessary - Heart valves	- Dissecting of a 4 chambered heart and examining of a model of a heart - Describing the action of heart muscles and movement of blood - Discussing the importance of heart valves	- Models - Mammalian heart - Recommended textbooks - ICT tools

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	relate the structure of blood vessels to their function	Arteries, veins and capillaries	Examining photomicrographs of blood vessels to compare their structures	- Biosets, bioviewers - Microscopes - Prepared slides - ICT tools
	relate pressure differences in the dual circulatory system to the functions of the two circuits	Pulmonary and systemic	Comparing pulmonary and systemic circulations	Recommended textbooks
	- identify possible causes of high blood pressure - identify possible causes of coronary heart disease - describe the effects of high blood pressure and coronary heart disease	- Genetic predisposition, stress, lack of exercise and diet - Diet, stress, smoking, obesity and high blood pressure - Stroke, heart attack and death	- Discussing possible causes of high blood pressure - Discussing possible causes of coronary heart disease - Discussing effects of high blood pressure and coronary heart disease	- Recommended textbooks - Resource person - ICT tools - Statistics showing the link between life style and occurrence of coronary heart diseases - Print media

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.6.10 Blood	state components of mammalian blood and their functions	- Red blood cells - White blood cells - Platelets - Plasma	Identifying red and white blood cells from prepared slides and photomicrographs	- Slides, photomicrographs - Recommended textbooks
	describe the movement of materials between capillaries and tissue fluid	Diffusion and pressure filtration	Discussing movement of materials between capillaries and tissue fluid by diffusion and pressure filtration	
7.6.11 Immunity	explain how the body protects itself against infection	Skin, tears, clotting of blood, white blood cells (engulfing action and production of antibodies)	Discussing natural defense mechanisms NB. Brief mention of movement of white blood cells through capillary walls	Print media
	describe events leading to active immunity	- Infection leading to antibody production - Vaccination	Discussing natural active immunity and artificial active immunity	Print media
	Describe events leading to passive immunity	Transfer via placenta and breast milk,	Discussing natural passive immunity in infants and artificial passive immunity	- Print media - Recommended textbooks

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
		serum injection (snake bites and rabbies)		
	describe the effects of the human immunodeficiency virus (HIV) on the body	- destruction of white blood cells - Inability to resist infections	- Discussing the effects of HIV on the body - Discussing the signs and symptoms of AIDS	Recommended textbooks
	discuss the significance of immunization	Extended programme of immunization	studying the immunization card	- Immunization card - Resource persons
7.6.11 Reproduction in Humans	describe the structure and function of the human reproductive system	Male and female reproductive systems	- Identifying parts of the reproductive system - Drawing and labelling the reproductive systems - Discussing the functions of parts of the reproductive system	- Print media - Recommended textbooks - ICT tools
	- state that meiosis results in formation of gametes - explain that gametes contain half set of chromosomes	- Gamete formation - Sperm and ovum NB. Details of meiosis not required	Discussing formation of gametes	Recommended textbooks

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	compare male and female gametes	Size, mobility and numbers produced	Comparing sperm and ovum	Print media
	describe the human menstrual cycle	- Changes in uterus lining and ovulation. Fertile and infertile phases of cycle - Role of oestrogen and progesterone	Describing the menstrual cycle	Print media
	outline pathway of sperm from testes to ovum	Pathway of sperm	Outlining the pathway of sperm	Print media
	describe fertilization	Fusion of gametes	Discussing fertilization	
	describe the early development of a zygote	- Mitotic cell division, multiplication, growth and development NB. Details of mitosis not required - Limited to the formation of a	Discussing early development of a zygote	Recommended textbooks

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
		ball of cells which becomes implanted in uterine wall Early development - Organ formation later development – mass increase		
	describe how the embryo is maintained and protected by the placenta	The placenta and umbilical cord: Passage of antibodies, exchange of nutrients, gases and excretory products. Separating foetal from maternal blood	Discussing the importance of the placenta and umbilical cord	- Print media - ICT tools

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	describe limitations of the placenta in protection of the embryo	Reference to passage of toxins and certain viruses including HIV	Discussing limitations of the placenta in protection of the embryo	
	describe how the embryo is protected in the uterus	- Amnion and amniotic fluid - Shock absorber - Preventing temperature fluctuation	Discussing the role of amnion and amniotic fluid	
	explain the dangers of taking drugs during pregnancy	Dangers of taking drugs during pregnancy (Illicit drugs, smoking and drinking alcohol)	Discussing the dangers of taking drugs during pregnancy	Resource person
	explain causes of infertility	Hormonal, sperm, quantity and quality, physical damage by	Discussing causes of infertility	Recommended textbooks

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
		STIs and cancers		
	outline various methods of contraception	- Abstinence and natural methods - Chemical: spermicides - Barriers: condom and diaphragm - Intra-uterine devices (IUDs) - Hormonal: injection and pills	Discussing reliability, limitations and appropriateness of contraceptive methods	- Resource person - Print media - Recommended textbooks
	- Identify causative agent of sexually transmitted infections (STI's) - describe symptoms and effects of STI's	- Causative agents - Gonorrhea, syphilis, chancroid and HIV/AIDS	- Discussing causative agents of STI's - Discussing effects on health and fertility by STIs	
	discuss the spread of STI's and HIV and the methods by which	Mutual faithfulness, long term relationships, abstinence	Discussing the spread of STI's and HIV, incurability of AIDS and the social implications	

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	they may be controlled	and the use of condoms • Contact tracing • Treatment with antibiotics for STI's		

7.8 TOPIC 8 MICROBIOLOGY AND BIOTECHNOLOGY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.8.1 Microbiology and Biotechnology	- list the main characteristics of the microorganisms - outline the economic importance of microorganisms	- Types of micro-organisms - Viruses - Bacteria - Fungi - Role of microorganisms in decomposition - Role of yeast in the production of bread and alcohol - Role of bacteria in yoghurt and cheese production - Role of the fungus <u>Penicillium</u> in the production of penicillin	- Observing structures of microorganisms on print media, microscope slides and ICT tools - Drawing and labelling the structures - Discussing the economic importance of microorganisms - Carrying out educational tours to bakeries, breweries and dairy industries	- ICT tools - Print media - Microscope slides - Bakeries - Breweries - Dairies

7.9 TOPIC 9 GENETICS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.9.1 Chromosomes and Genes	- Define chromosome - state that DNA is divided into sections called genes - define a gene as a unit of inheritance - distinguish between the terms gene and allele	- Chromosome as including a long molecule of DNA - Gene concept - Inheritance - Gene pairs (alleles)	- Discussing the - Structures of inheritance (chromosomes) - Units of inheritance (Genes) - Explaining genetic terms - Gene - Allele - Dominant - Recessive - Genotype - Phenotype	- ICT tools - Print media - Recommended textbooks
7.9.2 Monohybrid Inheritance	- describe complete dominance - predict possible outcomes of mono-hybrid crosses	- Dominant and recessive alleles - Phenotypes and genotypes - Genetic Crosses - Gametes F^1 and F^2 generations¹	- Discussing complete dominance - Constructing - Genetic crosses and punnet squares	- Recommended textbooks - ICT tools - Coloured beads/beans

			- Using coloured beads or beans picked at random to demonstrate inheritance	
	• explain codominance	• Inheritance of ABO Blood group • Phenotypes groups A, B, AB and O • Gene alleles for blood groups I^A, I^B, I^O • Sex chromosomes X and Y	• Constructing genetic crosses • Discussing and illustrating sex determination in mammals	• Print media
7.9.3 Mutations	• describe mutation • discuss factors that increase rate of mutation	• Genetic and chromosomal mutations • Mutation as a change in the structure of a gene or chromosomes or in the chromosome number - Sickle cell anemia - Albinism - Down syndrome • Factors that increase rate of mutation - Genetic disposition - Radiation (UV, Gamma, Alpha, X – rays) - Carcinogens	• Discussing the concept of mutation, types and factors that increase rate of mutation	• Print media • Recommended textbooks • ICT tools

7.10 TOPIC 10 BIODIVERSITY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.10.1 Biodiversity	- state the five kingdoms - state characteristic features of the five kingdoms	- Five kingdoms - Animalia - Plantae - Prokaryotae - Fungi - Protista - Diagnostic features	- Discussing characteristic features of each kingdom - Classifying organisms into five kingdoms	- Recommended textbooks - ICT tools - Samples of organisms

7.11 TOPIC 11 ECOSYSTEMS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.11.1 Ecosystems	- define ecosystem - list components of an ecosystem - identify soil as a key component of an ecosystem - identify biological components of soil	- A self-contained system of interdependent organisms and the physical environment - Physical and biological components such as air, water, soil, light and living organisms - Roles of soil - Litter, earthworms, nematodes, termites, fungi, bacteria and humus	- Examining an ecosystem such as pond, forest, field - Discussing the role of soil - Extracting macro-organisms in the soil - Demonstrating that soil contains micro-organisms	- Field trip - Wire gauze - Lamps - Soil samples - Alcohol - Lime water/bicarbonate indicator

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	- state the role of the biological components - compare properties of clay, loam and sandy soils	- Crumb structure, aeration, fertility and pH - Size of particles - Air content - Water holding capacity - Drainage - Leaching - Acidity/alkalinity (pH)	- Discussing the role of biological components - Carrying out experiments to show techniques of measuring air, water, organic content, pH of a soil sample, retention and drainage of clay and sandy soils	Soil samples
	describe a natural ecosystem	Natural ecosystems	Carrying out educational tours to a local ecosystem	Natural ecosystems
	construct simple food chains	Trophic levels: Producers, consumers and decomposers	- Discussing examples from local ecosystems - Constructing simple food chains and food webs	
	interpret food webs	Inter-relationships between food chains	Discussing inter relationships between food chains	Print media
	explain the loss of	Respiration and heat loss	Discussing energy losses in food chains	ICT tools

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	energy in food chains	Energy input and energy flow		
	Interpret ecological pyramids	Pyramids of numbers, biomass and energy	Constructing and interpreting pyramids	
	describe nutrient cycles	- Carbon cycle - Nitrogen cycle NB. Scientific names of bacteria not required	- Discussing nutrient cycles - Drawing nutrient cycles	Print media
	- describe an artificial ecosystem - compare species (bio) diversity in natural and artificial ecosystems - state problems caused by limited	- Artificial Ecosystem as human – influenced and with limited species diversity - Soil fertility - Pest problems	- Studying cultivated piece of land e.g. garden or field - Comparing species diversity - Discussing the problems	Artificial ecosystems

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	species diversity			

7.12 TOPIC 12 HEALTH AND DISEASES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.12.1 Health	- define health - discuss levels of hygiene	- Health as a state of physical, mental, emotional and social wellbeing and not just the absence of disease - Personal hygiene - Domestic hygiene - Community hygiene - Waste disposal methods - Provision of clean, safe drinking water - Sanitation - Provision of health facilities such as clinics	- Discussing the concept of health - Carrying out cleaning campaigns around communities and in school - Inviting resource persons to address pupils on health issues	- Resource persons - Cleaning materials - Recommended textbooks - ICT tools
7.12.2 Diseases	define disease	Disease defined as pathological condition of a body part, an organ or a system resulting from various causes such as:	Discussing the concept of disease and meaning of infectious and non-infectious diseases	- Recommended textbooks - ICT tools

		infection, genetic defect or environmental stress and characterized by an identifiable group of signs or symptoms		
	• explain with examples the meaning of infectious and non-infectious diseases • state the causative agent, mode of transmission, symptoms and ways of treating and preventing infectious diseases	• Types of diseases • Infectious diseases - Cholera - Malaria - Ebola - Typhoid - Bilharzia - HIV/AIDS • Non-infectious diseases - Deficiency diseases - Genetic diseases - Cancer	• Researching on causes, mode of transmission, symptoms and ways of treating and preventing infectious diseases • Discussing research findings	• ICT tools • Resource persons • Recommended textbooks

FORM 4**7.1 TOPIC 1 INTRODUCTION TO BIOLOGY**

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.1.1 Safety (labels and symbols)	• identify various safety labels and symbols • explain the meaning of various safety labels and symbols	• Labels and symbols	• Identifying various safety labels and symbols • Discussing the meaning of the various labels and symbols	• Containers with labels and symbols

7.2 TOPIC 2 CHEMICALS OF LIFE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
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7.2.1 Carbohydrates	• classify carbohydrates into monosaccharides, disaccharides and polysaccharides • state the uses of glucose, sucrose, starch, glycogen and cellulose in living organisms • describe the chemical structure of glucose	• Classification of carbohydrates as mono, di-, and poly-saccharides (glucose, sucrose, starch, glycogen and cellulose as examples) • Chemical structure of glucose • Uses of glucose, sucrose, starch, glycogen and cellulose in living organisms	• Classifying carbohydrates into monosaccharides, disaccharides and polysaccharides • Discussing uses of mono -, di- and polysaccharides • Illustrating the chemical structure of glucose • Making a model of a glucose molecule	• Recommended textbooks • ICT tools • Print media • Plasticine • Tooth picks • Cellotape
7.2.2 Proteins	• describe the chemical structure of an amino acid • state the functions of proteins in living organisms	• Chemical structure of an amino acid • Functions of proteins in living organisms	• Illustrating the chemical structure of amino acid • Discussing the functions of proteins in living organisms	• Recommended textbooks • Print media • ICT tools
7.2.3 Fats	• state the functions of fats in living organisms	• Functions of fats in living organisms	• Discussing the functions of fats in living organisms	• Recommended textbooks
7.2.4 Nucleic acids	• State the functions of nucleic acids in living organisms	• Functions of DNA and RNA	• Discussing the functions of DNA and RNA	• Recommended textbooks • ICT tools

7.3 TOPIC 3 ENZYMES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
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7.4.1 Enzymes	explain the use of enzymes in the production of washing powders	- Application of enzymes in industries - Detergents/washing powder - Lipases – fat stains - Proteases – blood stains	- Discussing the uses of lipases and proteases in stain removal - Carrying out experiments with lipases and proteases to remove stains	- Washing powder - Lipases and proteases - Recommended textbooks
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7.5 TOPIC 4 PLANT SCIENCE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.5.1 Reproduction	- define asexual reproduction - describe natural and artificial methods of vegetative reproduction in plants	- Asexual reproduction in plants - Vegetative reproduction - Natural methods - Rhizomes, tubers, suckers, runners, bulbs - Artificial methods - Cuttings, grafting	- Discussing asexual reproduction in plants - Observing vegetative structures in plants - Demonstrating various methods of vegetative reproduction	- Recommended textbooks - Plant vegetative structures such as: stems, tubers, leaves, bulbs - Educational tours
	- state advantages and disadvantages of vegetative reproduction - describe the structure and	- Resistance to disease, genetic variation, survival of offspring, rate of propagation - Sexual reproduction in plants	- Discussing advantages and disadvantages of vegetative reproduction - Observing different types of flowers using hand lenses	- Grass - Irish potatoes - Bananas - Sugar cane - Sweet-potato stems - Onion bulbs - Strawberry plants - Flowers - Hand lenses

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	characteristics of wind and insect pollinated flowers • compare the different structural adaptations of insect pollinated and wind pollinated flowers • outline the process of pollination • distinguish between self-and cross pollination	• Flower as a reproductive structure in plants • Wind pollinated and insect pollinated flowers • Pollination	• Examining pollen grains using microscopes or hand lenses • Comparing using fresh specimens, an insect –pollinated and a wind pollinated flower	• Microscopes • Pollen grains
	• outline the process of fertilization	• Development of pollen tube • Fusion of male and female nuclei NB. No knowledge of double fertilization is required	• Practical examination of growing pollen tubes • Describing the growth of the pollen tube and its entry into the ovule followed by fertilization	• Print media • ICT tools
	• describe changes that occur after fertilization in a flower	• Formation of seeds and fruits	• Collecting seeds/fruits • Observing a variety of fruits and seeds	• Fruits • Seeds • Hand lenses

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	- describe seed and fruit dispersal - outline the importance of seed/fruit dispersal	- Seed and fruit dispersal - Importance of seed dispersal - Colonization of new areas - Reducing overcrowding of plants - Increasing survival chances of plants	- Relating structure of seed/fruit to method of dispersal - Preserving collected seeds (project)	- Manila - Plastics - Cellotapes - Disinfectants
	- discuss advantages and disadvantages of sexual reproduction in plants - describe the basic structure of a fruit - describe the structure of a monocot and dicot seed - distinguish between monocot and dicot seeds - describe the environmental	- Advantages and disadvantages of sexual reproduction in plants - Fruit and seed structure - Fruit – fruit wall (pericarp seeds, locule scars and remains of style and sepals) - Seed – Testa (seed coat) embryo (radicle and plumule) cotyledon, endosperm, hilum micropyle - Monocot seeds – maize seeds - Dicot seeds – bean seeds - Germination	- Discussing advantages and disadvantages of sexual reproduction in plants - Observing a variety of fruits and seeds - Identifying different parts (external and internal) of fruits/seeds) - Making labelled drawings of seeds and fruits - Carrying out controlled experiments to investigate	- Recommended textbooks - ICT tools - Variety of fruits and seeds - Scalpel blades - Hand lenses - Bean seeds - Maize seeds - Pyrogalllic acid

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	conditions necessary for germination • describe the functions of enzymes in the germination of seeds	• Factors affecting germination - Oxygen - Suitable temperature - Water Functions of enzymes - Respiration - Hydrolysis of stored food	conditions necessary for germination	• Drying agents such as calcium chloride • Refrigerator • Incubator
7.5.2 Coordination and Response	• define phototropism and geotropism • illustrate phototropism and geotropism in plants • name the plant hormones • describe the effects of the hormones in plant growth and development	• Phototropism • Geotropism • Plant hormones as chemical substances produced in one part of a plant and transported to other parts where they initiate a response - Absciscic acid - Ethylene - Gibberellins - Auxin - Cytokinins • Effects of plant hormones	• Carrying and experiments to illustrate phototropism and geotropism • Discussing causes of response of plants to light and gravity • Carrying out experiments to illustrate effects of plant hormone ethylene • Discussing commercial application of plant hormones	• Potted plants • Germinating seeds • Cardboard boxes • Blades • ICT tools • Petri-dishes • Cotton wool • Clinostat • Plant hormones • Fruits • Young plants

7.6 TOPIC 5 ANIMAL SCIENCE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.6.1 Productivity	- define food conversion efficiency - explain productivity - identify animal pests and related diseases - describe how animal pests and diseases affect productivity	- Food conversion efficiency - Productivity - Animal pests and diseases - Anthrax – bacteria - Foot and mouth – virus - Liver damage – fluke - Red water – tick (vector)	- Measuring the mass of an animal - Constructing and interpreting growth curves - Discussing productivity - Discussing prevention mechanisms	- Rabbit and chicken - Resource persons - Recommended textbooks
7.6.2 Homeostasis	- define homeostasis - Identify and label parts of the skin - Relate the functions of parts of the skin to the maintenance of a constant body temperature	- The maintenance of a constant internal environment - Skin structure and functions of the parts - Temperature sensitivity, shivering, sweating, vaso-dilation and vasoconstriction	- Identifying substances and conditions that are regulated such as water, pH, ion, glucose concentration and temperature - Identifying the parts and discussing their functions - Relating parts functions to temperature regulation	- Print media - Recommended textbooks

	describe the coordination function of the brain in homeostasis	- The hypothalamus - Negative feedback	Discussing the role of the hypothalamus in temperature regulation (negative feedback)	- Recommended textbooks - Print media
	define excretion	Excretion as removal of toxic materials and waste products of metabolism	Discussing excretion	Recommended textbooks
	list substances that must be excreted and organs involved	- Kidney – urea, salt, water and toxins - Lungs – carbon dioxide and water - Skin – salts and water	Discussing the excretory organs and the wastes they excrete	
	- describe the structure of the urinary system - state the functions of the parts of the urinary system - describe the structure of the kidney - outline the functions of the kidney	- Structure of the urinary systems - Functions of kidney, urethra, ureter, bladder - Kidney structure - Functions - urine formation, osmo-regulation, pH regulation (Ultrafiltration and selective reabsorption in outline only) NB. No detailed structure of the nephron required	- Drawing and labelling the urinary system - Discussing functions of parts of the urinary system - Discussing kidney structure and functions	Print media
	Describe how the kidney dialysis machine works	Kidney failure and the use of dialysis machine	Discussing how the dialysis machine works	Educational tour
7.6.4 Coordination and Control	describe the structure of a neurone	Structure of sensory relay and motor neurons	Drawing and labelling the structure of neurons	- Print media - Bioviewer and biosets

	• describe a simple spinal reflex • outline functions of the main parts of the brain	• Receptor, sensory neuron, relay neuron, motor neuron, effector and spinal cord • Cerebrum, cerebellum, pituitary gland, hypothalamus and medulla	• Demonstrating reflex actions such as knee jerk and blinking • Drawing the reflex arc • Identifying parts of the brain and discussing their functions	• Print media • Bioviewer • Print media
	• describe the gross structure of the eye • describe the functions of parts of the eye • describe how the eye focuses images on the retina • Describe eye defects and their correction	• Eye • Front view and horizontal section • Parts of the eye • Refraction, accommodation and image formation • Short and long sightedness	• Observing own eye using a mirror • Drawing and labelling the front view and horizontal section of the eye • Discussing eye defects and their correction	• Mirror • Model of an eye • Lenses
7.6.4 Endocrine System	• define hormone • describe action of a hormone • describe the effects of adrenaline • outline the role of insulin and glucagon in homeostasis	• Hormone • Chemical substance produced by a gland and carried by blood, which affects the activities of a target organ • Production, transport and effect • Adrenal gland, fight or tight responses • Regulation of blood glucose level • Negative feedback mechanism	• Discussing the production, transport and effect of hormones (insulin and adrenaline) • Discussing regulation of blood sugar	• Recommended textbooks - Recommended textbooks - Resource person

	state the signs of diabetes mellitus	Diabetes mellitus signs: frequent urination, tiredness, thirst, increased blood glucose level and glucose in urine	Discussing signs of diabetes mellitus	
7.6.5 Skeletal system	outline functions of the skeleton	Support, protection, movement and locomotion	- Observing the human skeleton - Discussing functions of the skeleton	Model of the skeleton
	- Identify the main bones of the fore and hind limb of a mammal - describe types of joints - describe the types of movement permitted by joints - describe the action of muscles at a hinge joint	- Scapula, humerus, radius ulna, pelvis femur, tibia and fibula - Hinge joints, ball and socket joint - Hinge joint – one plane - Ball and socket joint – all planes - Antagonistic muscles extensor and flexor muscles	- Identifying the bones on the skeleton - Observing joints on the model of a skeleton - demonstrating movement permitted by joints - Demonstrating extensor and flexor muscles	- Model of antagonistic muscles - Pupils

7.7 TOPIC 6 RESPIRATION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.7 Respiration	define aerobic respiration	Aerobic respiration	Discussing aerobic respiration	Recommended textbooks

	• identify mitochondria as the site of aerobic respiration	• Mitochondria and aerobic respiration	• Identifying mitochondria as the site for respiration	
	• state the equation for aerobic respiration • list uses of energy in the body	• Word equation • Metabolic processes	• Carrying out experiments to show release of heat and carbon dioxide and oxygen uptake • Discussing uses of energy in the body	• Food sample
	• define anaerobic respiration • state the equations for anaerobic respiration in plants and animals • describe the production and effects of lactic acid on muscles during exercise	• Anaerobic respiration in plants and animals • Fatigue • Oxygen debt • Role of liver	• Carrying out experiments to show Fermentation of sugar by yeast • Discussing effects of lactic acid on muscle during exercise	• Fermentation • Apparatus - Recommended textbooks

7.8 TOPIC 7 MICROBIOLOGY AND BIOTECHNOLOGY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.8.1 Recombinant Gene Technology	• define the term recombinant DNA	• Recombinant DNA as a DNA molecule incorporating DNA from	• Discussing formation of recombinant DNA	• ICT tools

		more than one species of organism		
	• explain that genes may be transferred between cells of different species	• Production of insulin using recombinant DNA technology • Gene coding for human insulin inserted into DNA of bacteria (recombinant DNA) • Transformed bacteria allowed to multiply producing insulin • Insulin then recovered from bacteria NB. Only simplified outline of insulin production using recombinant DNA technology	• Discussing insulin production	• Print media • Recommended textbooks
	• discuss potential benefits and hazards of recombinant DNA technology	• Benefits and hazards of recombinant DNA technology	• Debating on benefits and hazards of recombinant DNA technology	

7.9 TOPIC 8 GENETICS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.9.1 Variation	• identify variable features in plants and animals	• Variation - Features such as: leaf size, size of pods,	• Collecting leaves of the same plant and measuring length	• Leaves • Balances • Rulers

	- distinguish between continuous and discontinuous variation - draw graphs to show continuous distribution of characteristics	height, coat colour, mass, tongue rolling, sex - Continuous and discontinuous variation - Tables and graphs	- Measuring pupils' heights/masses in class - Counting number of boys and girls in class - Constructing tables and graphs to show continuous and discontinuous variation	Graph papers
	identify causes of variation	Genetic and environmental causes	Discussing causes of variation in plants and animals	- Recommended textbooks - ICT tools
7.9.2 Selection	- explain how selection occurs in natural communities - define evolution - discuss natural selection as a possible mechanism for evolution	- Variation, competition, selective survival and reproduction - Evolution - Natural selection - Struggle for existence and survival of the fittest	Discussing how selection occurs in natural communities	Recommended textbooks
	- describe the applications of artificial selection - give reasons for plant and animal breeding	- Application of artificial selection in plants and animal breeding - Productivity, quality of breed, resistance to drought and disease	Investigating application of artificial selection in plant and animal breeding in communities and commercial farms	- Local communities - Commercial farms

7.10 TOPIC 9 BIODIVERSITY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.10.1 Biodiversity	identify threats to biodiversity	- Threats to biodiversity - Deforestation - Invasive species	- Identifying threats to biodiversity - Discussing threats to biodiversity	Recommended textbooks

	• explain threats to biodiversity	- Habitat destruction - Climate change		
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7.11 TOPIC 10 ECOSYSTEMS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.11.1 Management of Ecosystem	• describe effect of ground cover on run off, erosion and evaporation • describe the harmful effects of humans on ecosystems • explain the concept of carrying capacity	• Top soil preservation, water retention by soil, role of mulching as artificial cover • Agricultural, industrial and social activities resulting in climate change. Soil erosion pollution, desertification and reduction in biodiversity • Carrying capacity • Limiting factors: - Water - Food - Oxygen - Space and shelter	• Investigating effects of soil cover on run-off, erosion and evaporation • Carrying out case studies of environmental problems in the local communities • Suggesting and implementing corrective measures • Discussing the concept of carrying capacity	• Recommended textbooks • Print media • Local ecosystems • Recommended textbooks

	describe the effects of exceeding the carrying capacity	Degradation of ecosystem; overgrazing, deterioration of veld, disease and pollution	Discussing the effects of exceeding the carrying capacity	Education tours
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7.12 TOPIC 11 HEALTH AND DISEASES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.12.1 Drug use and Abuse	- define drug - describe medical uses of drugs	- Drug as any externally administered substance that modifies or affects chemical reactions in the body - Antibiotics – penicillin - Analgesics – paracetamol – aspirin - Anti-malaria – chloroquine – paludrine NB. Refer to allergic reactions	- Discussing the meaning of drug - Discussing the importance of proper administration and use of medicinal drugs. - Discussing allergic reactions to drugs in some individuals	- Recommended textbooks - ICT tools

	• describe the effects of alcohol, tobacco, smoking, cannabis and solvents	• Physical, mental and social ill-effects - Alcohol - Tobacco - Cannabis - Solvents	• Discussing the effects of drug abuse • Carrying out campaigns against drug abuse	• Resource persons • Print media
	• Describe uses of electricity in the home • Calculate cost of electricity • Discuss different methods of saving electricity • Explain the use of solar photo voltaic systems	• Heating • Lighting • Powering electrical devices • Meter reading costs per unit • Energy saving bulbs, solar panels, biogas • Alternative sources of energy, switch off appliances not in use • low power rating appliances • solar photo voltaic cells	• Discussing uses of electricity in the home • Reading of electricity meter, determining the costs of electricity used • Discussing ways of saving electricity • Discussing the use of solar systems as alternative sources of energy in homes	• Electrical appliances used in the home e.g. electric jug, iron, welding machines, TV, cake mixers • Energy saving bulbs, other bulbs • Photo voltaic cells

BIOLOGY SYLLABUS F3-4 2024 - 2030

9.0 ASSESSMENT

The Lower Secondary Biology syllabus learning area for Forms 3 – 4 shall be assessed through School Based Continuous Assessment (SBCA) and Summative Assessment (SA). These assessments shall be guided by the principles of inclusivity, practicability, authenticity, transparency, flexibility, validity and reliability. The principles are crucial for creating a supportive and effective learning environment that fosters growth and development in learners at secondary school level. Arrangements, accommodations and modifications shall be visible to enable candidates with special needs to access assessments.

This section covers the assessment objectives, the assessment model, the scheme of assessment, and the specification grid.

9.1 Assessment Objectives

By the end of the Lower Secondary Biology syllabus learning area for Forms 3 - 4, learners will be assessed on their ability to:

- 9.1.1 Show knowledge and understanding.
- 9.1.2 Handle information and solve problems.
- 9.1.3 Display experimental skills and investigations.

9.2 Assessment Model

Assessment of learners at Lower Secondary school level for Biology Syllabus shall be both Continuous and Summative as illustrated in Figure 1. School Based Continuous Assessment shall include recorded activities from the School Based Projects. The mark shall be included on the learner's end of term and year report. Summative assessment at school level shall include terminal examinations which are at the end of the term and year.

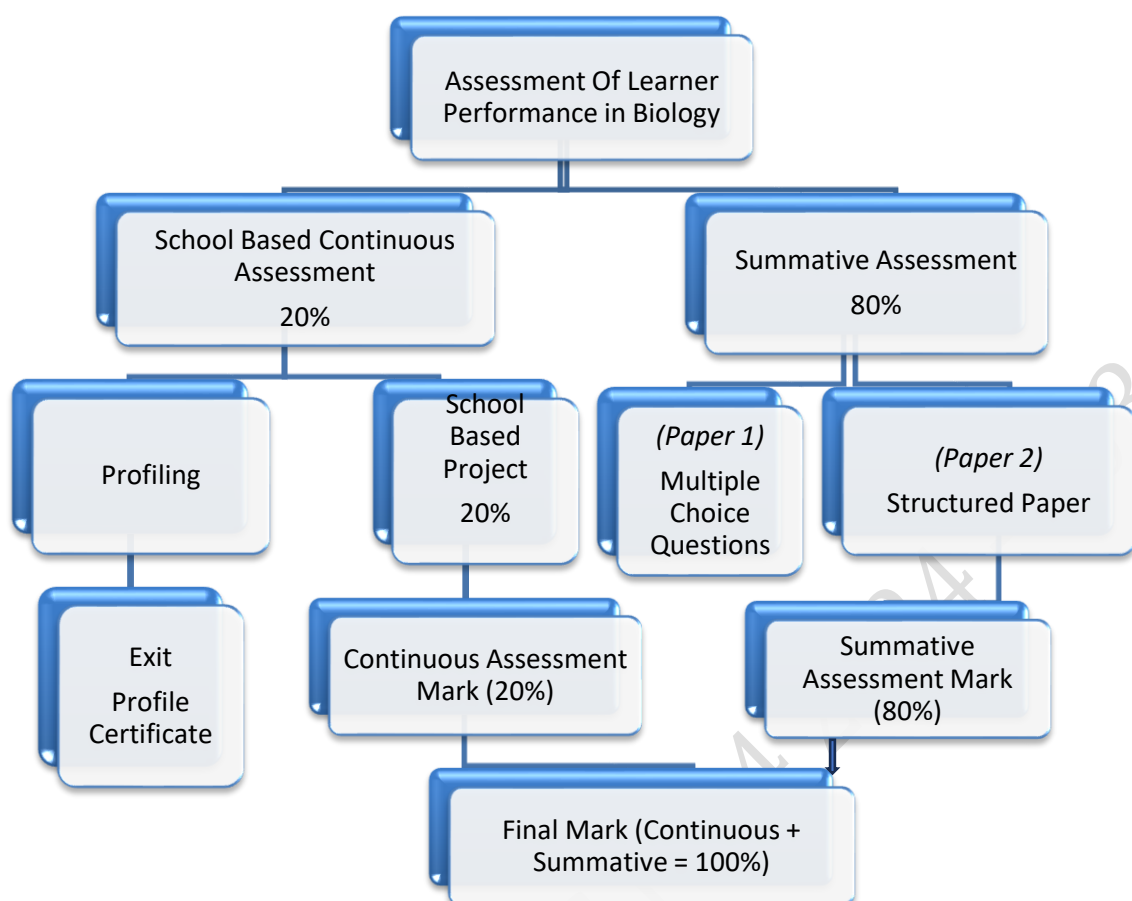


Fig. 1 Assessment Model

In addition, learners shall be profiled and learner profile records established.

Learner profile certificates shall be issued for checkpoints assessment in schools as per the dictates of the Teacher's Guide to Learning and Assessment. The aspects to be profiled shall include learner's prior knowledge, values and skills, and subsequently the new competences acquired at any given point.

9.3 Scheme of Assessment

The Assessment Model shows that learners shall be assessed using both School Based Continuous Assessment and Summative Assessment for both School and ZIMSEC assessments.

The table shows the Scheme of Assessment where 20% is allocated to School Based Continuous Assessment and 80% to School or ZIMSEC Summative Assessment.

FORM OF ASSESSMENT	WEIGHTING
School Based Continuous Assessment	20%
Summative Assessment	80%
Total	100%

9.3.1 Description of School Based Continuous Assessment

Learners shall do one school-based project per Form which contributes to 20% of the end of year final mark. The end of year summative assessment shall then contribute 80%. However, for ZIMSEC public examinations, two (2) school-based projects shall be considered as School Based Continuous Assessment at Form 4. The two School Based Projects shall include those done during Form 3. and Form 4 sessions. Each will contribute 10%.

9.3.1.1: School – Based Project Continuous Assessment Scheme

The Table given below shows the Learning and Assessment Scheme for the School Based Project.

Project Execution Stages	Project Stage Description	Timelines	Marks
1	Problem Identification	January	5
2	Investigation of related ideas to the problem/innovation	February	10
3	Generation of possible solutions	March	10
4	Selecting the most suitable solution	April-May	5
5	Refinement of selected solution	June	5
6	Presentation of the final solution	July	10
7	Evaluation of the solution and Recommendations	August-September	5
	TOTAL		50

The learning and assessment scheme shows the stages that shall be executed by pupils and the timeline at which each stage shall be carried out. Possible marks, totalling 50, are highlighted to indicate how much can be allocated.

9.3.2 Description of the ZIMSEC Summative Assessment

ZIMSEC Summative Assessment shall be a public examination at Form 4. The examination shall consist of three (3) papers of different weighting

Paper	Paper type	Marks	Duration	Weighting
1	Multiple choice	40	1hr	18%
2	Theory	100	2hrs 15mins	44%
3	Practical test	40	2hrs	18%
TOTAL				80%

Paper 1: Multiple choice

Duration: 1 hour

The paper consists of 40 compulsory multiple-choice items of the direct choice type. Each question shall have 4 response items.

Paper 2: Theory

Duration: 2 hours 15minutes.

The paper has 2 sections

Section A will carry 40 marks and will consists of number of compulsory structured questions of variable mark valve.

Section B carries 60 marks and will consist of 4 structured questions. Each question will carry 20 marks. Candidates will be required to answer any 3 questions.

Paper 3: Practical Test

Duration: 2hours

This paper will consist of 2 compulsory questions each carrying 20 marks.

Paper 4: Continuous Assessment

This paper will consist of 2 projects during the 6 terms.

9.4 Specification Grid

SKILL	Paper 1	Paper 2	Paper 3
KNOWLEDGE AND UNDERSTANDING	15%	15%	
ANALYSIS, SYNTHESIS AND EVALUATION	45%	45%	
APPLICATION AND PROBLEM SOLVING	40%	40%	
TOTAL	100%	100%	100%

BIOLOGY SYLLABUS F3-4 2024 - 2030

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